

Task 5: Capstone Project & Incident Response – Theory Report

1. Introduction

Cybersecurity incidents such as unauthorized access, malware infections, and data breaches are increasing rapidly in today's digital environment. Organizations must not only identify vulnerabilities but also respond effectively to security incidents.

This capstone project focuses on applying core cybersecurity concepts through vulnerability assessment, exploitation (controlled), monitoring, and incident response simulation in a controlled lab environment.

2. Objective

The primary objectives of this task are:

To apply cybersecurity skills in a self-chosen capstone project.

To identify vulnerabilities in a target system or application.

To analyze security logs and detect suspicious activities.

To simulate an incident response lifecycle including detection, containment, eradication, and recovery.

To document findings with evidence and propose mitigation strategies.

3. Scope of the Project

The scope of this project includes:

Vulnerability assessment and penetration testing on a deliberately vulnerable system.

Log analysis for detecting security incidents.

Simulation of an incident response process.

Preparation of professional documentation and presentation.

Out of Scope:

Attacking real-world systems without authorization.

Performing destructive attacks on production environments.

4. Tools and Technologies Used

Kali Linux – For reconnaissance, scanning, and exploitation.

Transplantable 2 / DVWA / Vulnerable Test Network – Target system for testing.

Nmap – Network scanning and service enumeration.

Nikto – Web server vulnerability scanning.

Metasploit Framework – Exploitation and post-exploitation.

ELK Stack (Optional) – Log analysis and monitoring.

Wireshark (Optional) – Network traffic analysis.

5. Project Planning

Project planning involved:

Defining project goals and timeline.

Selecting a vulnerable target environment.

Identifying attack vectors and response strategies.

Designing a basic network diagram showing attacker and target systems.

6. Implementation Methodology

6.1 Reconnaissance & Scanning

Identified live hosts and open ports using Nmap.

Enumerated services and operating system details.

Scanned web applications for known vulnerabilities.

6.2 Vulnerability Assessment

Identified misconfigurations, outdated services, and weak authentication.

Assessed severity and potential impact of discovered vulnerabilities.

6.3 Exploitation (Controlled)

Exploited selected vulnerabilities using Metasploit.

Gained limited shell access to demonstrate risk.

Ensured all actions were performed ethically in a lab environment.

6.4 Evidence Collection

Collected screenshots of scans, exploit results, and system access.

Logged commands and outputs for documentation.

7. Incident Response Simulation

7.1 Detection

Detected suspicious activities through logs and alerts.

Identified abnormal login attempts and exploit patterns.

7.2 Containment

Isolated the affected system to prevent further compromise.

Disabled vulnerable services temporarily.

7.3 Eradication

Removed malicious sessions and payloads.

Patched vulnerable services and updated configurations.

7.4 Recovery

Restored services securely.

Verified system integrity and normal operations.

8. Findings

Multiple vulnerabilities were found due to outdated software and weak security configurations.

Lack of monitoring can delay incident detection.

Simple vulnerabilities can lead to full system compromise if not addressed.

9. Mitigation Strategies

Regular patching and updates.

Strong authentication mechanisms.

Network segmentation and firewall rules.

Continuous monitoring and log analysis.

Periodic vulnerability assessments

10. Conclusion

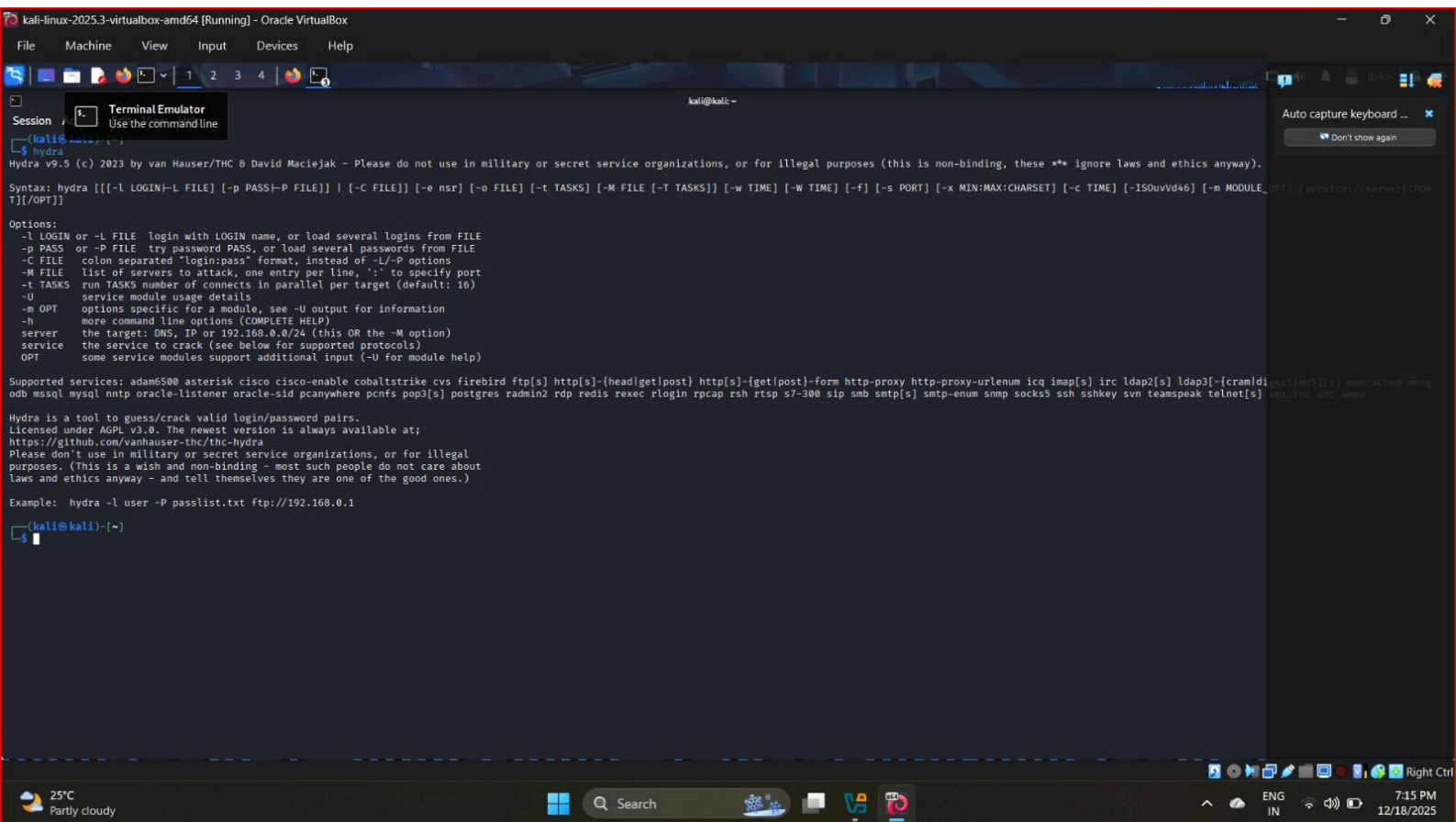
This capstone project provided hands-on experience in vulnerability assessment and incident response. It demonstrated how cyber attacks occur and how structured incident response procedures help reduce damage and restore systems securely. The project enhanced practical understanding of ethical hacking and defensive security mechanisms.

```
Sorry, try again.
[sudo] password for msfadmin:
Sorry, try again.
sudo: 3 incorrect password attempts
msfadmin@metasploitable:~$ sudo service mysql start
[sudo] password for msfadmin:
Sorry, try again.
[sudo] password for msfadmin:
Sorry, try again.
[sudo] password for msfadmin:
sudo: 3 incorrect password attempts
msfadmin@metasploitable:~$ service apache2 start
The program 'service' can be found in the following packages:
 * debian-helper-scripts
 * sysuconfig
Try: sudo apt-get install <selected package>
-bash: service: command not found
msfadmin@metasploitable:~$ service mysql start
The program 'service' can be found in the following packages:
 * debian-helper-scripts
 * sysuconfig
Try: sudo apt-get install <selected package>
-bash: service: command not found
msfadmin@metasploitable:~$
```

```
* sysvconfig
Try: sudo apt-get install <selected package>
-bash: service: command not found
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:9c:c4:ed
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.0
          inet6 addr: fd17:625c:f037:2:a00:27ff:fe9c:c4ed/64 Scope:Global
          inet6 addr: fe80::a00:27ff:fe9c:c4ed/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:38  errors:0  dropped:0  overruns:0  frame:0
          TX packets:80  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4922 (4.8 KB)  TX bytes:11096 (10.8 KB)
          Base address:0xd020  Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:142  errors:0  dropped:0  overruns:0  frame:0
          TX packets:142  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0 txqueuelen:0
          RX bytes:44105 (43.0 KB)  TX bytes:44105 (43.0 KB)

msfadmin@metasploitable:~$ _
```



```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali:~$ hydra
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway)

Syntax: hydra [[-l LOGIN|-L FILE] [-p PASS|-P FILE]] [-c FILE] [-e nsr] [-o FILE] [-t TASKS] [-M FILE [-T TASKS]] [-w TIME] [-W TIME] [-f] [-s PORT] [-x MIN:MAX:CHARSET] [-c TIME] [-ISOuvVd46] [-m MODU
T][[/OPT]]

Options:
-l LOGIN or -L FILE login with LOGIN name, or load several logins from FILE
-p PASS or -P FILE try password PASS, or load several passwords from FILE
-c FILE colon separated "login:pass" format, instead of -L/-P options
-M FILE list of servers to attack, one entry per line, ':' to specify port
-t TASKS run TASKS number of connects in parallel per target (default: 16)
-U service module usage details
-m OPT options specific for a module, see -U output for information
-h more command line options (COMPLETE HELP)
server the target: DNS, IP or 192.168.0.0/24 (this OR the -M option)
service the service to crack (see below for supported protocols)
OPT some service modules support additional input (-U for module help)

Supported services: adam6500 asterisk cisco cisco-enable cobaltstrike cvs firebird ftp[s] http[s]-{head|get|post} http[s]-{get|post}-form http-proxy http-proxy-urlenum icq imap[s] irc ldap2[s] ldap3[-{cra
odb mssql mysql nntp oracle-listener oracle-sid pcanynwhere pcnfs pop3[s] postgres radmin2 rdp redis rexec rlogin rpcap rsh rtsp s7-300 sip smb smtp[s] smtp-enum snmp socks5 ssh sshkey svn teamspeak telnet

Hydra is a tool to guess/crack valid login/password pairs.
Licensed under AGPL v3.0. The newest version is always available at;
https://github.com/vanhauser-thc/thc-hydra
Please don't use in military or secret service organizations, or for illegal
purposes. (This is a wish and non-binding - most such people do not care about
laws and ethics anyway - and tell themselves they are one of the good ones.)

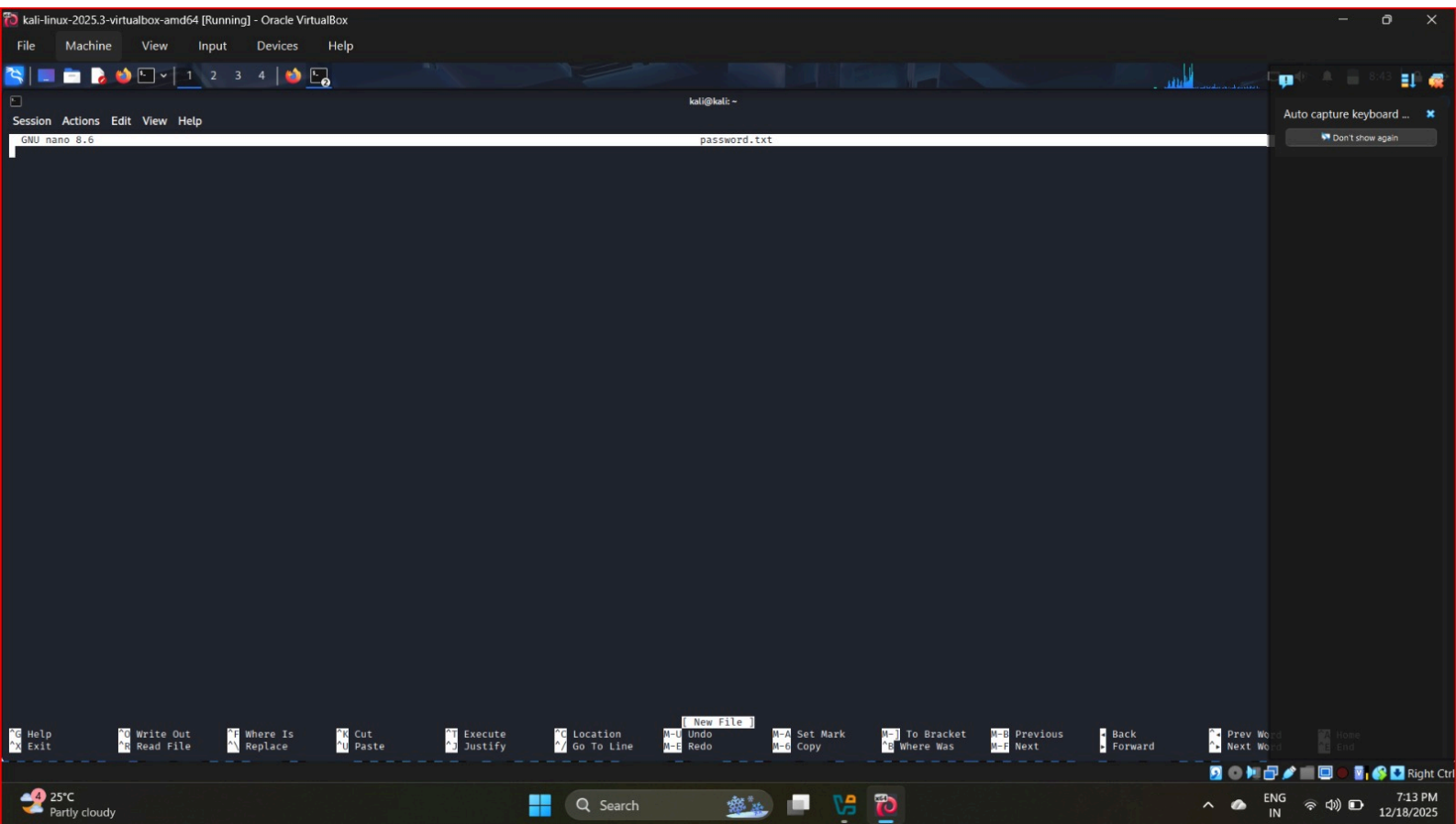
Example: hydra -l user -P passlist.txt ftp://192.168.0.1

kali@kali:~$ hydra -l admin -P passwords.txt
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway)

hydra: invalid option -- '1'

kali@kali:~$ $ <TARGET-IP> https-post-form .
zsh: no such file or directory: TARGET-IP

kali@kali:~$
```



kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

about:sessionrestore x Login - Damn Vulnerable x +

127.0.0.1:42001/login.php

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

Auto capture keyboard ...
Don't show again



Username
admin

Password
.....

Login

[Damn Vulnerable Web Application \(DVWA\)](#)

23°C
Partly cloudy

Search

ENG
IN

10:53 PM
12/18/2025

